

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Class :SYIT	Batch : 01
Date of Assignment :05-08-2026	Date/Time of Submission :05-08-2026

Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

Code:

```
DataStructurePrac7A.py - C:/Users/mvlui/OneDrive/Document
File Edit Format Run Options Window Help

size = 5
table = [ [] for _ in range(size)]

def insert(key, value):
    table[ (key % size) ].append ((key, value))

insert(10, "Apple")
insert(15, "Banana")
insert(20, "Mango")
insert(32, "Guava")
insert(33, "Pineapple")

print(table)

print("NirajRane_S032")
|
```

Output:

```
== RESTART: C:/Users/mvlui/OneDrive/Documents/SYIT_032/DataStructurePrac7A.py ==
[(10, 'Apple'), (15, 'Banana'), (20, 'Mango'), [], [(32, 'Guava'), (33, 'Pineapple')], []]
NirajRane_S032
|
```

b. Store and retrieve data from the hash table.

Code:

```
DataStructurePrac7B.py - C:/Users/mvlui/OneDrive/Documents/SYI
File Edit Format Run Options Window Help

size = 5
table = [ [] for _ in range(size)]

def insert(key, value):
    table[(key % size)].append((key, value))

def search(key):
    for k, v in table[key%size]:
        if k==key:
            return v
    return "Not Found"

insert(10, "Apple")
insert(15, "Banana")
insert(20, "Mango")
insert(32, "Guava")
insert(33, "Pineapple")

print(table)

key = int(input("Enter Key :"))
print(search(key))

print("NirajRane_S032")
```

Output:

```
===== RESTART: C:/Users/mvlui/OneDrive
[[ (10, 'Apple'), (15, 'Banana'), (20, 'Mango') ], [], [(32, 'Guava'), (33, 'Pineapple')], []]
Enter Key :32
Guava
NirajRane_S032

===== RESTART: C:/Users/mvlui/OneDrive
[[ (10, 'Apple'), (15, 'Banana'), (20, 'Mango') ], [], [(32, 'Guava'), (33, 'Pineapple')], []]
Enter Key :15
Banana
NirajRane_S032

===== RESTART: C:/Users/mvlui/OneDrive
[[ (10, 'Apple'), (15, 'Banana'), (20, 'Mango') ], [], [(32, 'Guava'), (33, 'Pineapple')], []]
Enter Key :67
Not Found
NirajRane_S032
```